Problem Statements - Ciphathon 26'

Note : You are free to come-up with your own solutions for Expected Outcomes

CIPH-PS-001: AI in Cybersecurity

Problem Statement Title: Local Semantic Phishing Detector (Soft-Power)
Problem Statement Description:
Nation-state actors use "no-payload" text-based social engineering to bypass traditional filters. Detecting these in air-gapped, high-privacy environments without cloud-based AI remains a critical gap.
Expected Outcome
Build a local AI engine that identifies malicious intent by analyzing linguistic deviations, unauthorized hierarchy leverage (Soft-Power), and suspicious "channel-shift" requests within purely text-based communications.

CIPH-PS-002: AI in Cybersecurity

Problem Statement Title: Mitigation and Detection of Prompt Injection Attacks in Large Language Models
Problem Statement Description:
Large Language Models (LLMs) remain highly susceptible to prompt injection vulnerabilities.Current defenses often struggle to distinguish between complex, legitimate user queries and malicious "jailbreak" attempts; consequently, this creates a significant security gap where attackers can manipulate the model into performing unauthorized actions, generating harmful content, or compromising the integrity of the entire AI-integrated system.
Expected Outcome:
A simple security tool that checks user input before it reaches the AI.The final result should be a safer AI that can block attacks without slowing down for regular users.

CIPH-PS-003:AI in Cybersecurity

Problem Statement Title: Virtual Security Officer and Intelligent Cyber Intelligence Platform
Problem Statement Description:
Small and medium-sized businesses often lack the budget for full-time security experts, which leaves them vulnerable to hidden technical risks. These risks include expired SSL certificates, leaked subCIPH-PS-001s, and poorly configured email security like SPF or DKIM records. While automated scanners can find these issues, the data is usually too technical for a business owner to understand or act upon quickly.
Expected Outcome:
A working security dashboard that automatically checks CIPH-PS-001 health and server configurations. This platform must feature a conversational AI agent that explains risks in simple business terms. The system should provide a live security maturity score that updates as fixes are applied; additionally, it must be able to handle "crisis events" by alerting the user and providing a step-by-step guide to resolve issues like CIPH-PS-001 reputation drops or exposed sensitive assets.

CIPH-PS-004:Blockchain & Web3 Security

Problem Statement Title: Blockchain-Based Federal Identity Verification
Problem Statement Description:
Current federal identity systems rely on centralized databases that create single points of failure. These servers are highly vulnerable to large-scale data breaches and identity theft. Because different agencies often use incompatible systems, users may face administrative friction and a lack of control over their own personal data; furthermore, the process of verifying a user's identity currently requires sharing sensitive raw information that can be easily intercepted or misused.
Expected Outcome:
A blockchain-based system that issues secure, encrypted digital identities to users; this platform must allow agencies to verify a user's status instantly without storing their private information in a central database. The final solution should provide a user-friendly interface for users to manage their own data; additionally, it must use cryptographic proofs to confirm identity details securely, reducing the risk of fraud and eliminating the need for repetitive manual paperwork.

CIPH-PS-005: Blockchain & Web3 Security

Problem Statement Title: Automated Smart Contract Vulnerability Scanner
Problem Statement Description:
Many developers deploy smart contracts to the blockchain without performing a full security audit; This leads to common but devastating exploits such as re-entrancy attacks, where a malicious contract drains funds by repeatedly calling a function before the first call finishes. Other frequent issues include weak access controls that allow unauthorized users to take over the contract and logic flaws that can freeze assets forever. Because blockchain transactions are permanent and cannot be reversed, identifying these security gaps before deployment is essential; however, manual code review is often too slow or expensive for smaller projects.
Expected Outcome:
A functional scanning tool that analyzes smart contract code to flag known vulnerabilities. This tool must provide a clear report that explains why a specific line of code is risky. The system should categorize findings by severity-such as "Critical" for potential fund loss or "Low" for gas optimization-and suggest simple code fixes for each identified issue.

CIPH-PS-006: Cybercrime Awareness & Digital Safety

Problem Statement Title: Dynamic Behavioral Safety Scoring for Real-Time Digital Hygiene
Problem Statement Description: Traditional security assessments are static "one-time" quizzes that quickly become obsolete as users adopt new technologies. Consequently, individuals often develop a false sense of security by relying on basic tools like VPNs while remaining blind to modern risks. This "Identity Debt" accumulates through over-permissive OAuth grants, shadow AI accounts, and session token vulnerabilities that can bypass Multi-Factor Authentication (MFA). Currently, there is no automated system to track these live risks or provide a "Live Security Score" for digital hygiene; this lack of visibility makes it difficult for users to understand how their daily digital behaviors, such as linking apps to their work email or leaving sessions active on public devices, directly impact their overall safety posture.
Expected Outcome:A diagnostic tool that aggregates live data from account permissions, app-layer vulnerabilities, and historical breach databases to generate a dynamic "Digital Safety Posture Score"; this platform must provide real-time "nudges" or alerts that suggest specific actions to fix high-risk behaviors. The system should go beyond simple checklists by identifying complex threats like “Session hijacking” or "Shadow AI" usage; finally, it must offer a visual dashboard that shows how the user’s score improves as they revoke unnecessary permissions or secure their digital identity.

CIPH-PS-007: Cybercrime Awareness & Digital Safety

Problem Statement Title: Personal Digital Privacy and Exposure Intelligence Engine
Problem Statement Description:
Most individuals lack a clear understanding of their "Digital footprint" and how fragmented pieces of information across the web can be combined to target them. For instance, a username used on a developer forum can often be linked to a social media profile, an old data breach, or a public code repository. Currently, there is no easy way for a non-technical person to see a consolidated view of what the internet "knows" about them; consequently, users remain overexposed and unaware of hidden risks like leaked metadata in photos or historical credential patterns that have appeared on the dark web.
Expected Outcome:
A user-friendly intelligence engine that takes simple identifiers,such as an email address or username,and generates a detailed report of a person's online visibility across social media, forums, and public databases; the system must quantify this exposure into a "Digital Risk Score" that is easy to understand. The platform should identify specific dangers like exposed personal location data or reused passwords found in old leaks; additionally, it must provide a dashboard with clear, actionable steps to help users delete old accounts, tighten privacy settings, or remove sensitive metadata to shrink their overall attack surface.

CIPH-PS-008: Cybercrime Awareness and Digital Safety

Problem Statement Title: Proactive Mobile Defense Against Evolving Scam Ecosystems
Problem Statement Description:
Modern fraud has moved beyond simple spam calls into complex, coordinated attacks like "Digital Arrest" scams, call bombing, and malicious APK distribution; these sophisticated social engineering campaigns often bypass traditional keyword filters by using realistic voice impersonation and high-pressure psychological tactics.There is a critical need for a mobile-first solution that can reason about fraud patterns as they happen, identifying the behavioral signs of a scam rather than just looking for known "Bad" phone numbers or words.
Expected Outcome:
A mobile application capable of ingesting voice transcripts, URLs, and file metadata to detect fraud in real time. The system must classify scams by their specific type,such as impersonation or phishing,and explain to the user exactly why a particular interaction is risky. The final platform should provide immediate protective guidance, like blocking a malicious download or suggesting the user hang up on a suspected "Digital Arrest" call; additionally, the app should use pattern correlation to identify if a single scammer is using multiple methods, such as following an SMS link with a coordinated phone call, to target the victim.

CIPH-PS-009: Privacy-First Applications

Problem Statement Title: Browser-Level Sensitive Data Leakage Prevention from LLMs
Problem Statement Description:
As more people use Large Language Models (LLMs) like ChatGPT or Claude for work, they often accidentally paste sensitive information, such as private API keys, internal company documents, or personal customer data, into the chat prompt. This creates a major security risk because that data is then stored and potentially used to train future AI models. Currently, there is no built-in protection in most web browsers to stop this information from leaving the user's computer; consequently, once a user clicks "send," the data is leaked, and there is often no way to delete it from the AI's memory.
Expected Outcome:
Lightweight browser extension that automatically monitors text boxes on popular AI websites. This tool must use pattern matching and local machine learning to detect sensitive data types like credit card numbers, passwords, and proprietary code snippets. The system should provide a clear warning to the user or "redact" the sensitive parts of the text before submission; additionally, it must operate entirely on the user's device to ensure that the extension itself does not become a new source of data leakage.

CIPH-PS-0010: Privacy-First Applications

Problem Statement Title: Granular and Consent-Centric Data Collection Flow
Problem Statement Description:
Most applications collect excessive user data without providing clear choices or meaningful control. This often leads to "privacy-by-design" being ignored in favor of bulk data harvesting. Users are typically forced into an "all-or-nothing" agreement where they cannot opt out of specific tracking behaviors; consequently, sensitive information is often gathered without the user's explicit understanding or genuine permission.
Expected Outcome:
A functional application flow that enforces granular, explicit consent for every data category. The system must allow users to toggle specific permissions on or off at any time. The final result should include a transparent audit log showing when and how consent was given; additionally, the app should be programmed to only function with the minimum data required for the user's selected features.

CIPH-PS-0011: Secure Web & API Design

Problem Statement Title: Autonomous API Discovery and Adaptive Threat Protection
Problem Statement Description: Many systems contain "zombie" or undocumented APIs that are no longer maintained, providing easy entry points for attackers to bypass security; additionally, a lack of real-time monitoring makes it difficult to distinguish legitimate traffic from malicious bots performing web scraping or brute-force attacks. Without a way to automatically map these hidden endpoints, sensitive data can be drained slowly and system resources overwhelmed before a breach is even detected. There is a need for an offline-first platform that discovers every API by analyzing source code and traffic, ensuring that no "shadow" services remain unmonitored or exposed to exploitation.
Expected Outcome:
An offline-first platform that creates a real-time inventory of all active and hidden APIs while using behavioral analytics to flag abnormal usage. The system must automatically trigger defenses like adaptive rate limiting or IP blocking when it detects suspicious patterns. The final solution should provide a dashboard visualizing traffic and risk levels; furthermore, it must demonstrate the ability to stop coordinated attacks without ever exposing internal source code to the internet.

CIPH-PS-0012: Secure Web & API Design

Problem Statement Title: Autonomous Secure Software Intelligence and Assurance Platform
Problem Statement Description: Traditional vulnerability scanners often produce long lists of disconnected bugs that fail to explain the actual business risk or how multiple weaknesses could be chained together for a major exploit. An intelligent, reasoning-driven engine is required to ingest source code and infrastructure signals to build a unified risk model; this system must go beyond simple scanning by providing a verifiable audit trail and an executive-level summary of security health.
Expected Outcome:
Participants must deliver a functional Autonomous Secure Software Intelligence & Assurance Platform that performs static code analysis and integrates runtime and infrastructure signals (such as SSL/TLS posture, exposed headers, and endpoint metadata) to construct a unified, multi-layer security risk model. The system should identify and correlate vulnerabilities into probability-weighted exploit chains, generate explainable findings with evidence traces and remediation guidance, and produce a structured executive-level “Virtual CISO” summary that translates technical risks into business impact and compliance alignment (e.g., OWASP categories). The platform must maintain a tamper-evident audit trail by cryptographically signing all generated reports and logs, enabling integrity verification and traceability. A working demonstration should showcase analysis of a sample application, visualization of correlated attack paths, prioritized remediation insights, and verifiable security intelligence outputs.

CIPH-PS-0013: SOC Automation & Incident Response

Problem Statement Title: Intelligent Network Defense and Deception Appliance
Problem Statement Description:
Standard firewalls are often reactive and easily bypassed by sophisticated attackers who use port scanning or lateral movement to navigate a network undetected. Furthermore, once an attacker is inside, there are few tools to study their behavior without risking real assets. An intelligent defense appliance is needed at the network edge to act as both a shield and a trap; this system should monitor traffic patterns while deploying "honey ports" to trick attackers into revealing their tactics.
Expected Outcome:
A deployable network appliance that functions as an adaptive firewall and a deception engine. The system must dynamically block threats like bot activity and brute-force attempts while misdirecting attackers toward decoy services. The final product should include a centralized dashboard with attack heatmaps and automated containment decisions; additionally, it must support secure VPN tunneling and traffic segmentation to keep the core network isolated.

CIPH-PS-0014: SOC Automation & Incident Response

Problem Statement Title: Automated Incident Triage and Prioritization Engine
Problem Statement Description:
Security Operations Center (SOC) teams are often overwhelmed by a high volume of low-value alerts; this "alert fatigue" leads to manual triage delays, which can allow a high-severity threat to go unnoticed for hours. There is a critical need for an automated logic layer that can instantly categorize and prioritize incidents based on their potential impact; this allows human analysts to focus their energy on the most dangerous attacks first.
Expected Outcome:
An automation engine that classifies incoming security alerts by severity and accuracy. The system should use pre-defined logic to filter out "false positives" and escalate critical incidents immediately. The demonstration must show a significant reduction in triage time compared to manual processing; furthermore, the tool should provide a clear reason for each priority score to help analysts make faster, more accurate decisions.

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